

QUESTION PAPER - SET 4

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If

9. yes, share details.

10. Tell us about your family.

11. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

In the late 21st century, the Global Council enacted the "Resource Equilibrium Accord" to address escalating conflicts over dwindling freshwater reserves. The Accord introduced a tiered quota system, assigning water access rights based on each nation's ecological footprint, technological capability for desalination, and per-capita consumption history.

While wealthier nations invested in advanced desalination plants and atmospheric water harvesters, poorer nations — despite having smaller ecological footprints — struggled to meet their citizens' basic needs under their reduced quotas.

Critics argued that the Accord unintentionally entrenched inequality, as historical consumption patterns favored nations that had previously overused water.

To address the backlash, an amendment proposed the creation of a "Water Equity Fund" financed by a surcharge on nations exceeding their quota. The fund would support sustainable water projects in under-resourced regions.

Yet, debates continued over enforcement: how to monitor usage accurately, prevent data manipulation, and ensure that the fund genuinely benefited those in need.

1. What was the main goal behind the introduction of the Resource Equilibrium Accord in the late 21st century?

- A. To establish a global water trading market
- B. To balance global water distribution and prevent conflicts over freshwater resources
- C. To reward nations with low per-capita water consumption
- D. To increase funding for desalination research

2. Which aspect most contributed to the Resource Equilibrium Accord unintentionally reinforcing inequality among nations?

- A. Technological advancements in desalination
- B. Inaccurate monitoring of water usage
- C. Historical consumption patterns that benefited countries with prior high water use
- D. Opposition to the Water Equity Fund

3. Which conclusion is best supported by the information provided in the passage about the Resource Equilibrium Accord?

- A. The amendment eliminated all concerns about fairness
- B. Nations with smaller ecological footprints were given larger quotas
- C. Wealthier nations opposed atmospheric water harvesting
- D. The Accord unintentionally reinforced pre-existing inequalities

4. Under the Resource Equilibrium Accord, when would a nation be required to pay a surcharge to support the Water Equity Fund?

- A. When its per-capita consumption is below the global average

- B. When it fails to reduce its ecological footprint
- C. When it does not participate in atmospheric water harvesting
- D. When its water usage exceeds the assigned quota

5. What is the main challenge that remains unresolved regarding the amendment to the Resource Equilibrium Accord?

- A. Lack of sustainable water project ideas
- B. The unwillingness of wealthier nations to pay surcharges
- C. Uncertainty in the fund's revenue source
- D. Ensuring transparency and preventing misuse of funds

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. Which of the following is true about software testing?

- a) Testing only focuses on functional requirements
- b) Software testing guarantees that the software is bug-free
- c) Software testing is an ongoing activity throughout the development lifecycle
- d) Software testing is not needed after deployment

2. What is Unit Testing?

- a) Testing individual components in isolation
- b) Testing the system's performance
- c) Testing database interactions
- d) Testing the user interface

3. What is a bug in software testing?

- a) A testing method
- b) A feature of the software
- c) A malfunction or flaw in the software
- d) A software update

SECTION 5 - NUMERICAL APTITUDE

1. What is Regression Testing?

- a) Testing only new features
- b) Conducting user interface checks
- c) Validating database constraints
- d) Re-testing existing code to ensure new changes don't affect functionality

2. What is the role of a test manager?

- a) To write code for the software
- b) To deploy the software to production
- c) To develop the test environment
- d) To oversee the testing process and ensure test cases are executed

3. What is User Acceptance Testing (UAT)?

- a) Debugging system errors
- b) Running automated test execution
- c) Validating business requirements with end-users
- d) Testing system architecture

SECTION 6 - MARITIME KNOWLEDGE

1. At which location on Earth does the Coriolis effect reach its greatest strength?

- A. At the Equator
- B. At the Tropics
- C. At 30° latitude
- D. At the Poles

2. According to ship safety regulations, which of the following must be made weathertight as part of the Conditions of Assignment?

- A. Hatchways situated on the freeboard deck
- B. Sounding pipes on deck
- C. Cargo hold ventilators below deck
- D. Scuppers above deck

3. What does the term “lee shore” refer to in seamanship?

- A. A shore protected from wind
- B. A shore facing the wind
- C. A dangerous shore onto which wind blows
- D. A shallow coastal area

4. Which of the following corrections must be applied to a sextant reading to accurately determine the observed altitude (Ho) of a celestial body?

- A. Index error, dip, refraction, semi-diameter & parallax
- B. Index error and dip only
- C. Refraction only
- D. Dip and refraction only

5. What is the primary purpose of a marine sextant?

- A. Measuring wind direction

- B. Determining celestial altitude
- C. Measuring sea depth
- D. Determining engine power

6. The observer's prime vertical passes through the east and west points of which of the following?

- A. Rational horizon
- B. Equinoctial
- C. Celestial meridian
- D. Great circle

7. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+5.5)?

- A. 13th April 2014, 1000 hrs LT
- B. 13th April 2014, 0430 hrs LT
- C. 13th April 2014, 1530 hrs LT
- D. 13th April 2014, 1200 hrs LT

8. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

9. Which instrument is used to measure the salinity of seawater?

- A. Hygrometer
- B. Sextant
- C. Salinometer
- D. Barograph

10. Which document contains detailed information about cargo carried on a vessel?

- A. Deck Log Book
- B. Cargo Manifest
- C. Engine Log Book
- D. Stability Booklet

11. What is the purpose of ballast water in a ship?

- A. To clean cargo tanks
- B. To improve ship stability and trim
- C. To increase engine efficiency
- D. To cool cargo holds

12. Which chart projection is most commonly used for marine navigation?

- A. Gnomonic Projection
- B. Mercator Projection
- C. Polar Projection
- D. Cylindrical Equal Area Projection

13. What does “abandon ship” mean?

- A. Leaving cargo operations
- B. Evacuating the vessel during emergency
- C. Docking at a port
- D. Entering restricted waters

14. Which method is most commonly used to determine compass error during sunrise or sunset?

- A. Latitude by meridian passage
- B. Azimuth
- C. Sextant altitude
- D. Amplitude

15. On a nautical chart, which color is commonly used to indicate shallow water and the boundaries of certain regulated areas?

- A. Gold
- B. Blue
- C. Black
- D. Green

16. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?

- A. Cold front
- B. Occluded front
- C. Warm front
- D. Stationary front

17. Which factor causes the settling error of a gyrocompass to become more pronounced?

- A. Operating at higher latitudes
- B. Operating at lower latitudes
- C. Increasing the vessel's speed
- D. Navigating in areas of high magnetic variation

18. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker
- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

19. What is the full form of CSK?

- A. Chennai Super Kings
- B. DDDD
- C. SSSS

D. FFFF

20. Which factor mainly affects ocean tides?

- A. Wind speed
- B. Earthquake activity
- C. Gravitational pull of the Moon and Sun
- D. Atmospheric pressure only

21. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

22. What is the main purpose of a ship's anchor?

- A. To increase vessel speed
- B. To hold the vessel in position
- C. To improve fuel efficiency
- D. To reduce rolling motion

23. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

- A. Anticlockwise and inward
- B. Clockwise and inward
- C. Clockwise and outward
- D. Anticlockwise and outward

24. The vertical distance between the upper edge of the deck line and the centre of the load line disc represents the:

- A. Summer Freeboard
- B. Summer Draft
- C. Fresh Water Allowance

D. Winter North Atlantic Freeboard

25. Which safety equipment is used to prevent drowning during emergencies?

- A. Compass
- B. Lifebuoy
- C. Sextant
- D. Radar reflector

26. Which part of a ship experiences the greatest bending moment in heavy seas?

- A. Midship section
- B. Forecastle
- C. Stern peak
- D. Bridge deck

27. What is the primary purpose of a ship's freeboard?

- A. To increase cargo capacity
- B. To improve engine efficiency
- C. To provide reserve buoyancy and safety
- D. To reduce fuel consumption

28. Which opening below is NOT considered while assessing Conditions of Assignment?

- A. Hatchway on freeboard deck
- B. Door in superstructure end bulkhead
- C. Sounding pipe opening in engine room
- D. Ventilator on freeboard deck

29. What is the SI unit used to measure atmospheric pressure?

- A. Knot
- B. Pascal
- C. Joule

D. Newton

30. Which gas is primarily responsible for the greenhouse effect?

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Helium